Effects of different types of exercise training followed by detraining on endothelium-dependent dilation in patients with recent myocardial infarct...

BACKGROUND: In coronary artery disease, exercise training (ET) is associated with an improvement in endothelial function, but little is known about the relative effect of different types of training. The purpose of this study was to prospectively evaluate the effect of different types of ET on endothelial function in 209 patients after a first recent acute myocardial infarction. METHODS AND RESULTS: Endothelial function was evaluated before and after 4 weeks of different types of ET and after 1 month of detraining by measuring flow-mediated dilation and von Willebrand factor levels at baseline and after ET. Patients were randomized into 4 groups: group 1, aerobic ET (n=52); group 2, resistance training (n=54); group 3, resistance plus aerobic training (n=53); and group 4, no training (n=50). At baseline, flow-mediated dilation was 4.5+/-2.6% in group 1, 4.01+/-1.6% in group 2, 4.4+/-4% in group 3, and 4.3+/-2.3% in group 4 (P=NS). After ET, flow-mediated dilation increased to 9.9+/-2.5% in group 1, 10.1+/-2.6% in group 2, and 10.8+/-3% in group 3 (P<0.01 versus baseline for all groups); it also increased in group 4 but to a much lesser extent (to 5.1+/-2.5%; P<0.01 versus trained groups). The von Willebrand factor level after ET decreased by 16% (P<0.01) similarly in groups 1, 2, and 3 but remained unchanged in group 4. Detraining returned flow-mediated dilation to baseline levels (P<0.01 versus posttraining). CONCLUSIONS: In patients with recent acute myocardial infarction, ET was associated with improved endothelial function independently of the type of training, but this effect disappeared after 1 month of detraining.